Sahaj Bhadja

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Summary

Software engineer with 2+ years building production AI systems, agentic workflows, secure document pipelines, and full-stack applications using Python, React, TypeScript, cloud infrastructure, and customer-driven integrations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant AI workflows on Azure SQL and Blob Storage for 500+ users, translating stakeholder requirements into secure schemas and indexed queries that reduced dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI orchestration for 1,000+ monthly LLM jobs using explicit states, retries, and failure isolation, tripling throughput while documenting production behavior for technical and non-technical collaborators.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python services and React workflows integrating GPT-4o, Claude, Qwen-VL, and OCR to process 10,000+ daily text and handwritten submissions with 90%+ grading consistency at production scale.
- Designed secure PDF, DOCX, and ZIP ingestion with concurrent AWS RDS-backed processing for 1,000+ files per batch, reducing customer turnaround by 60% while maintaining stable APIs under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform Python microservices and 20+ automated PyTest and Mockito suites with Jenkins CI/CD, improving release reliability while supporting client integrations across Windows, Linux, and macOS environments.
- Optimized REST API and ETL integrations using OOP patterns, AWS Glue, S3, and SQL, cutting response times by 30% and improving data-pipeline efficiency by 68% for customer deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a customer-facing MERN platform with REST APIs, WebSockets, and eight modular Node.js services, supporting 2M+ synchronized records while translating operational requirements into reliable real-time workflows for distributed teams.
- Automated Docker-based testing and deployments through Jenkins CI/CD, reducing release time by 50% while improving implementation consistency, documentation, and issue resolution across frequent customer-facing product updates.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Gathered client requirements and designed Flask APIs for booking, scheduling, payments, and status workflows across 50,000+ records, presenting system architecture and 12+ integration points before implementation.
- Integrated Firebase synchronization, JSON webhooks, and automated payment workflows, reducing manual handling by 70% while improving troubleshooting visibility, implementation consistency, and communication around daily operational issues for customer operations.

Skills

AI & Agents: LLM orchestration, GPT-4o, Claude, Qwen-VL, LLaMA-2, OCR, RAG, embeddings, FAISS, Pinecone, prompt engineering, evaluation, human-in-the-loop workflows; **Engineering:** Python, TypeScript, JavaScript, React.js, Node.js, FastAPI, Flask, REST APIs, WebSockets, SQL, PostgreSQL, MySQL, Azure SQL, MongoDB; **Cloud & Delivery:** AWS, Azure, GCP, S3, AWS Glue, Docker, Kubernetes, Jenkins, Git, CI/CD, PyTest, Mockito, Agile/Scrum, debugging, documentation, customer integrations

Projects

Big-Brain Live | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an AI document teammate for 200+ users with secure uploads, semantic search, source-grounded summarization, and collaborative workflows, maintaining 95% query accuracy while reducing document-analysis time by 60%.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Node.js*
- Built an AI-powered database workspace supporting 10+ systems, combining schema-aware SQL generation, data exploration, and cross-platform deployment to accelerate customer workflow configuration across Windows, macOS, and Linux.

HMIS Live | *MERN, TypeScript, REST APIs, Sentry, Twilio*
- Implemented consent-based patient workflows, secure API integrations, automated notifications, and production monitoring, enabling 100+ daily appointments while reducing administrative workload by 30% across customer-facing healthcare operations and service delivery.